Tokens, Pixels, or Position? A Factorial Study of Why Vision-Language Models Misread Table Cells

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Abstract

When a financial page loses resolution, vision-language models (VLMs) misread the numbers on it. We ask what the failure consists of. On TAT-DQA documents we report interventional Shapley attribution of question-answering loss across image quality; pixel-level digit edits in real table cells; and a 2×2 factorial experiment that crosses a localization cue with restored glyph pixels at a fixed image-token count, on Qwen2.5-VL-7B, Qwen3-VL-8B and InternVL3-8B. Perception's share of the loss rises from 10–14% to 50–65%. Models read the digits actually drawn rather than the ones context suggests. Merely enlarging a downsampled page recovers much of the loss on the Qwen models, restored pixels add most at the lowest resolution, and a box helps mainly when digits are legible. Log-probabilities separate right from wrong readings while stated confidence does not. The confirmatory run stopped early (218, 260 and 127 of 500 planned cells) and our preregistered separability criterion was not met, so these findings are provisional.

1 Introduction

Answering a question about a financial table requires locating a cell, reading its digits, and computing with them. Benchmarks report the product of these steps. When accuracy falls on a degraded page, the score cannot say whether the model lost the cell, lost the digits, or never received enough of the image to see them.

These explanations imply different remedies. Lost position calls for a better search; lost pixels call for a better scan or a refusal; too few image tokens call for a resize before inference. Recent work has established parts of this picture. Shi (2026) enlarges rendered prose by bicubic interpolation and shows reading recover on Qwen3-VL-8B, and attributes InternVL3.5's insensitivity to its fixed tiling. Res-Bench separates information content from token length with padding and learned super-resolution (Li et al., 2025). Two studies report that red boxes give little benefit on compressed or many-page inputs (Shi, 2026; Xia et al., 2026). We did not find, in the work we reviewed, a design that crosses restored pixels with a localization cue at a matched token count on real table cells.

We contribute three things:

- An interventional Shapley curve of question-answering loss over image quality, attributing it to perception, computation and scale.
- A causal test by digit edit: one digit of a table cell is redrawn in pixels, and we measure whether the model reports the drawn number, the deleted one, or a placebo.
- A 2×2 factorial of localization and legibility at fixed image-token count, placed against the real downsampled image.

Our protocols were fixed before their main outputs existed, and we report where we departed from them. One departure matters throughout: the confirmatory experiment was stopped before its planned sample because the compute was needed elsewhere, and two interim looks preceded the stop (Section 6).

2 Related Work

Resolution and reading. Shi (2026) is the closest work. It enlarges the same image to several character heights, so that each size carries the same information. It measures a read-use gap on prose evidence, and also compares a box, copied pixels and enlargement as separate arms. Li et al. (2025) and Hu et al. (2026b) study resolution and token budgets more broadly, and Alonso and Lapata (2026) compresses tables under a token budget without upscaling images already below it. Lin et al. (2026) shows that small resizes crossing tiling or

token-grid boundaries can flip answers. Our addition is the comparison of enlarged and sharp pixels at one token count, on single cells of real tables.

Localization cues. Xia et al. (2026) finds that red boxes add little over a many-page context while crops recover most of the gain from extracted text. Our pages are single tables, and our box is crossed with legibility rather than tested alone.

Causal diagnosis. CADET decomposes tasks into unit tasks and scores prerequisites by necessity and sufficiency through prompt-level injection, without an image-quality axis (Hu et al., 2026a). Exact Shapley values have been applied to LLM agent components (Liu, 2026); we apply the value (Shapley, 1953) to VLM pipeline stages over image quality. Hwang et al. (2026) erases answer-relevant pixels and treats abstention as correct, whereas we replace a digit, so a faithful reader has a definite correct output.

Confidence. Token-level signals have been reported to exceed stated confidence for small VLMs under degradation (Ferdous, 2026b,a). The direction is not universal: in ConfBench the two models supporting both families disagree (Roy et al., 2026). Abstention is known to be rare (Madhusudhan et al., 2026). We do not offer this gap as a new phenomenon, only as a measurement on table cells along a resolution axis.

3 Attribution across Image Quality

Setup. We used 360 TAT-DQA training questions (Zhu et al., 2022), one per document, unused by any other experiment, on Qwen2.5-VL-7B (Bai et al., 2025b) and Qwen3-VL-8B (Bai et al., 2025a). Levels are s100, s60, s50, s40 and s30 (edge scale) and *photo* (s60 with small rotation, blur and JPEG compression). Three players repair a stage: P adds gold cell values to the prompt, C runs the model's expression in Python, and S supplies the gold scale. Scoring uses a port of the official TAT-QA exact-match metric (Zhu et al., 2021). Intervals are 95% document-bootstrap percentile intervals.

Results. Table 1 and Figure 1 show perception's share rising from 10% to 50% on 7B and from 14% to 65% on Qwen3. The locked primary criterion was met on both: the change in share is +39.7 [+33.0, +46.3] and +51.4 [+45.0, +58.5] points. At s100, computation and scale jointly exceed perception by +37.7 [+31.7, +44.3] and +34.6 [+28.3, +40.4] points, also meeting its criterion. On clean pages these models fail mostly after reading; at s30 reading is as large as, or larger than, everything else.

Model	Level	EM	ϕ_P	$\phi_C + \phi_S$	P share
Qwen2.5-VL-7B	s100	37.2	4.9	42.6	10%
	s50	26.4	11.0	47.9	19%
	s40	18.6	19.1	41.8	31%
	s30	15.3	30.3	30.2	50%
	photo	26.4	13.7	41.9	25%
Qwen3-VL-8B	s100	36.4	6.4	41.1	14%
	s50	27.5	10.2	39.8	20%
	s40	24.2	18.7	34.4	35%
	s30	6.7	43.3	23.4	65%
	photo	30.3	11.5	38.5	23%

Table 1: Shapley attribution of exact-match loss (points). $n = 360$ questions.

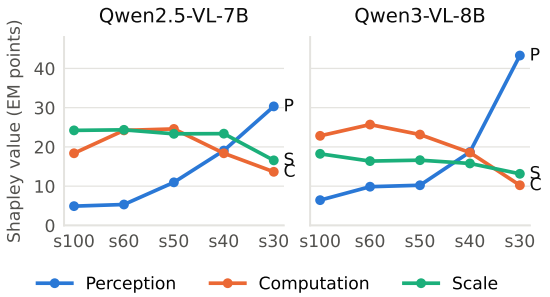


Figure 1: Shapley values by image quality. P, C and S mark the end of each series.

The P player is coarse. It bundles position and glyphs and, on span questions, nearly hands over the answer. The next two sections open it up.

4 Reading Experiments

Cells. The planned set has 500 target cells, one per document, disjoint from the attribution set: numeric, at least three digits, unique on the page, with row and column labels found through OCR. Models are the two above plus InternVL3-8B (Zhu et al., 2025) with tiling enabled, since the checkpoint default squeezes a page into one 448 px tile. Decoding is greedy. Prompts name the row and column labels and ask for the number. **The run stopped at 218, 260 and 127 complete cells**; the cells are taken in a fixed shuffled order, and the stop did not depend on results.

Digit edits. We delete the number in pixels using the modal background colour and redraw it in a matched font, in three versions: unchanged (null), one interior digit replaced (counterfactual, cf), and, as a placebo, a digit value never drawn. T is the

Criterion	7B	Qwen3	InternVL
Valid	yes	yes	yes
D1 separable	no	no	no
D2 context pull	negligible	negligible	negligible
D3 task dependence	open	open	yes
D4a hidden signal	yes	no	no
D4b verbal refusal	no	no	no
D4c signal–speech gap	yes	yes	yes

Table 2: Locked criteria on the incomplete sample. All are provisional. For D1, the entry lists qualifying levels.

Model	Lvl	T(cf)	$O - A$	T(copy)$-T$(expr)
7B	s50	70.6	+0.9	$+1.4$ [-2.3, +5.5]
	s40	49.5	+0.9	$+4.6$ [+0.0, +9.6]
	s30	26.1	+0.9	$+3.7$ [+0.0, +7.8]
Qwen3	s50	73.5	+1.2	$+1.2$ [-2.3, +4.2]
	s40	56.2	+0.0	-0.8 [-4.2, +2.7]
	s30	9.2	+0.4	$+1.5$ [-1.5, +5.0]
Intern	s50	80.3	+0.0	$+7.1$ [+1.6, +13.4]
	s40	78.7	+0.0	$+7.9$ [+0.8, +15.7]
	s30	59.1	+0.0	$+3.1$ [-3.9, +10.2]

Table 3: Digit edits. $T-$ columns are point differences with 95% intervals.

rate of outputting the drawn number, O the deleted original, A the placebo.

Factorial. All four conditions downsample the page and enlarge it back to its original size with bicubic interpolation, which equalizes image tokens. *Base* is that canvas. G pastes the original pixels back over every numeric OCR word on the page, excluding year-like words unless they are the target, so sharpness does not reveal the target and labels stay blurred. L draws a red box around the target and tells the model so. We report two-player Shapley values ϕ_L, ϕ_G on T and their interaction. *Real* is the actual downsampled page, with fewer tokens (on a 1240×1605 page, 7B receives about 285 image tokens at s30 and 2,508 at original size).

Confidence and refusal. We compare mean-token log-probability with a stated 0–100 confidence, and offer an UNREADABLE option under two wordings. Blank pages provide a floor.

Criteria. Table 2 applies the locked criteria. D1 requires both ϕ_L and ϕ_G to have intervals above zero at one level. D2 asks whether $O - A$ is at least 3 points with an interval above zero. D3 asks whether copying and writing an expression differ by at least 5 points. D4a requires log-probability AUROC of at least 0.75 at s40 and s30, above that of glyph height, with discordant within-cell pairs above 50%.[1] D4b requires a refusal gap of at least 20 points at s30. D4c requires log-probability AUROC to exceed that of stated confidence.

5 Results

Validity. At s100, the null edit shifts T by -1.4, -0.8 and +0.0 points; T(cf) is 86.7, 86.2 and 81.1; and a box alone directs copying to a decoy cell at 91.3, 95.8 and 85.0%. All three models pass, so the edits and the box function as intended.

[1]The protocol does not state the levels for the discordant-pair part. We apply the stricter reading, both levels.

Models read what is drawn. $O - A$ at s30 is +0.9 [+0.0, +2.3] on 7B and +0.4 [-0.8, +1.9] on Qwen3, and D2 classes the context pull as negligible on all three (Table 3). InternVL produced no original-digit outputs in this sample, so its interval is degenerate. The task effect is less uniform. It is inconclusive on the Qwen models and present on InternVL, where copying beats writing an expression by +7.1 [+1.6, +13.4] points at s50. That criterion compares output formats and does not establish a difference in perception.

The factorial. The locked separability criterion D1 was not met at any level on any model, so our preregistered stop rule for this part is triggered on the present sample. It was also unstable across looks: Qwen3 and InternVL each met it at s50 at a smaller interim sample, and ϕ_L at s40 on 7B has a lower bound at zero. The pattern is in Table 4. What follows is exploratory.

ϕ_L is largest at s50 (+5.7 [+0.7, +10.6] on 7B, +7.7 [+3.5, +12.1] on Qwen3), where restored pixels add almost nothing. ϕ_G is largest at s30 (+21.8 [+17.2, +26.6], +19.0 [+15.2, +22.9], +28.3 [+22.4, +34.6]), where ϕ_L cannot be told from zero. This crossover suggests that position limits reading at mild degradation and legibility at severe degradation. We chose this reading after seeing the data.

The interaction at s30 is large and positive (+10.6 [+3.2, +17.9], +25.8 [+19.6, +32.3], +29.9 [+20.5, +39.4]). On blurred pixels the box does not help at s30 and can harm: -3.7 [-11.5, +4.1], -16.9 [-23.5, -10.0], -18.1 [-26.8, -9.4]. On restored pixels it helps: +6.9 [+2.3, +11.9], +8.8 [+4.6, +13.5], +11.8 [+4.7, +19.7]. The harm may be an artifact, since a box can cover small blurred glyphs, and so ϕ_L may understate the value of position. Published box nulls (Shi, 2026; Xia et al., 2026) come from compressed pages and many-page contexts;

Model	Lvl	Real	Base	L	G	LG	ϕ_L / ϕ_G
7B	s50	80.7	86.7	90.4	85.3	93.1	+5.7* / +0.7
	s40	61.0	84.9	84.4	84.4	94.0	+4.6 / +4.6*
	s30	38.5	68.8	65.1	85.3	92.2	+1.6 / +21.8*
Qwen3	s50	78.1	86.5	93.8	87.3	95.4	+7.7* / +1.2
	s40	60.4	85.8	85.8	87.3	95.0	+3.8 / +5.4*
	s30	16.2	79.2	62.3	85.4	94.2	-4.0 / +19.0*
Intern	s50	81.9	82.7	84.3	81.1	90.6	+5.5 / +2.4*
	s40	80.3	80.3	76.4	79.5	90.6	+3.5 / +6.7*
	s30	62.2	63.0	44.9	76.4	88.2	-3.1 / +28.3*

Table 4: Percentage of cells read correctly. Real has fewer tokens on Qwen models; the other columns share a canvas. * marks intervals excluding zero.

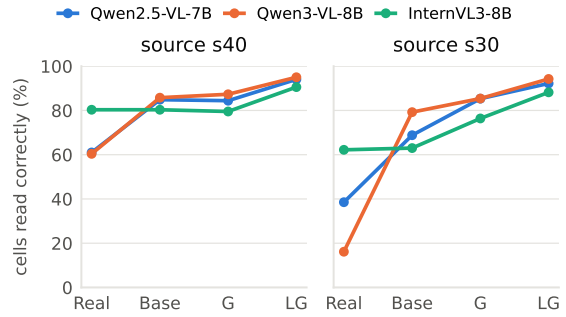


Figure 2: Reading rate from the real downsampled page through enlargement (Base), restored pixels (G), and box (LG).

ours concern a single table and may differ for that reason.

Enlargement. Enlarging adds no information, yet Base exceeds Real by +30.3 [+22.9, +37.6] points on 7B and +63.1 [+57.3, +68.8] on Qwen3 at s30 (Figure 2). On InternVL it adds +0.8 [+0.0, +2.4], as expected when both inputs become the same tile grid. This replicates on table cells what Shi (2026) shows on prose. Reporting this difference was planned, but the inference is post hoc, since we noticed it at an interim look. A locked follow-up on fresh cells (`protocol_token`) was stopped after too few cells to analyse, so its criteria remain untested.

For Qwen3 at s30, the steps are 16.2 (Real), 79.2 (Base), 85.4 (G), 94.2 (LG). Most of the attribution curve's perception loss at s30 on this model may therefore reflect the input size rather than destroyed information. This inference crosses experiments, one on question answering and one on copying, and we did not test it directly.

Confidence. Stated confidence carries almost no information (AUROC 0.51, 0.47, 0.47 at s30), and D4c holds on all three (Table 5). Excluding InternVL's few out-of-scale outputs leaves its AUROC at 0.48. Log-probability reaches 0.87, 0.85 and 0.79. The stronger D4a passes only on 7B; Qwen3's discordant-pair interval at s30 includes 50% (+61.2 [+49.0, +72.9]), and InternVL at s40 rests on few cells. So log-probability separates correct from incorrect readings across cells, while within-cell evidence is firm only on 7B.

Model	Lvl	Correct	AUROC$_{lp}$	AUROC$_{conf}$	Refusal	Refusal gap
7B	s40	61.0	0.82	0.50	6.0	+5.7
	s30	38.5	0.87	0.51	6.9	+11.2
Qwen3	s40	60.4	0.75	0.51	8.5	+10.1
	s30	16.2	0.85	0.47	16.5	+19.7
Intern	s40	80.3	0.76	0.45	9.4	+23.1
	s30	62.2	0.79	0.47	12.6	+13.2

Table 5: Log-probability vs stated confidence, and refusal under the stronger wording. Gap: P(refuse | wrong)−P(refuse | right), points.

Refusal. D4b fails on all three models: the `UNREADABLE` gap at s30 is +11.2 [+6.3, +16.9], +19.7 [+14.4, +25.2] and +13.2 [+0.6, +26.3] under the stronger wording. Qwen3 falls just short of the 20-point threshold. Silence is not absolute either. All three models refuse 100% of blank pages under that wording, and refuse 6.9, 16.5 and 12.6% at s30. Under the plain prompt, they output a number on 34.4, 49.2 and 54.3% of blank pages.

6 Limitations

Incomplete sample. We committed to conclude only from 500 cells and did not reach them: 44%, 52% and 25% of the plan. InternVL's estimates are especially wide. Two interim looks occurred, one requested and one during tooling tests, and verdicts changed between them. D1 and D4a in particular sit near boundaries. No threshold or scoring rule changed after looking. There is no multiplicity correction.

Unconfirmed explanation. Enlargement changes token count and glyph size per patch together, and interpolation may matter; our follow-up was to examine this and did not complete. We cannot claim a token-budget mechanism.

Unexplained disagreement. Shi (2026) reports a read-use gap on Qwen3-VL-8B even with gold evidence alone, whereas we find little task effect on Qwen models. Multi-hop prose answering and copying one cell differ, but we did not test whether that accounts for it.

Stimuli and scope. Downsampling is simulated and the box and pasted sharpness are artificial cues. Edits use substitute fonts. Our sets require facts matching OCR, and all data come from one domain. Attribution's P player is coarse. We tested three open models, and InternVL3-8B differs from the InternVL3.5 of prior work.

7 Conclusion

Perception's share of question-answering loss grows sharply as financial pages lose resolution, and models then read what is drawn rather than what context suggests. Our provisional data suggest this perception loss is not one thing: some may come from input size, some from destroyed glyphs, and some from position, with position and legibility trading places as degradation worsens. The preregistered separability test did not pass, so this structure is a hypothesis for the completed run, not a finding. Completing the planned sample and the token follow-up are the next steps, and a second dataset with real captures is needed before generalizing.

References

Iñigo Alonso and Mirella Lapata. 2026. A Table Is Worth 64 Tokens: Pixel-level Compression for Multi-Table Document Question Answering. *Preprint*, arXiv:2608.26949.

Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, Zesen Cheng, Lianghao Deng, Wei Ding, Chang Gao, Chunjiang Ge, Wenbin Ge, Zhifang Guo, Qidong Huang, Jie Huang, Fei Huang, Binyuan Hui, Shutong Jiang, Zhaohai Li, Mingsheng Li, and 45 others. 2025a. Qwen3-VL Technical Report. *Preprint*, arXiv:2511.21631.

Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, and 8 others. 2025b. Qwen2.5-VL Technical Report. *Preprint*, arXiv:2502.13923.

M M Asif Ferdous. 2026a. Bigger or Cheaper? Scale and Quantization Effects on Uncertainty Signals in Vision-Language Models Under Image Degradation. *Preprint*, arXiv:2607.24440.

M M Asif Ferdous. 2026b. Small Vision-Language Models Know When They Are Wrong But Cannot Say So: A Two-Model Study of Stated versus Internal Confidence Under Realistic Image Degradation. *Preprint*, arXiv:2607.22034.

Xia Hu, Brian Potetz, Chun-Ta Lu, Huanfen Yao, Leonidas Guibas, Zhicheng Wang, Howard Zhou, Pengfei Xing, and Andrew Gallagher. 2026a. Where MLLMs Fail and Why: Causal Task Decomposition for Capability Failure Diagnosis. *Preprint*, arXiv:2609.38851.

Zhensheng Hu, Guanglong Xu, and Guanqiang Hu. 2026b. The Answer Knows Where to Look: Zooming before MLLMs Answer under Visual-Token Budgets. *Preprint*, arXiv:2610.32664.

Taewook Hwang, Inbum Heo, Sung Jun Lee, and Sangkeun Jung. 2026. Perceptual hallucination in vision–language models: Definition, analysis and verification. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 24710–24725, San Diego, California, United States. Association for Computational Linguistics.

Chenxu Li, Zhicai Wang, Yuan Sheng, Xingyu Zhu, Yanbin Hao, and Xiang Wang. 2025. Res-Bench: Benchmarking the Robustness of Multimodal Large Language Models to Dynamic Resolution Input. *Preprint*, arXiv:2510.16926.

Dingyang Lin, Yingfeng Luo, Chenglong Wang, Chenwei Zhu, Anxiang Ma, Jingbo Zhu, and Tong Xiao. 2026. Reading Right, Answering Wrong: How Visual Configuration Changes Affect Evidence Use in VLMs. *Preprint*, arXiv:2609.25770.

Ming Liu. 2026. More Is Not Always Better: Cross-Component Interference in LLM Agent Scaffolding. *Preprint*, arXiv:2605.05716.

Nishanth Madhusudhan, Vikas Yadav, and Alexandre Lacoste. 2026. Knowing When Not to Answer: Evaluating Abstention in Multimodal Reasoning Systems. *Preprint*, arXiv:2604.14799.

Priyashree Roy, Sujitha Martin, Mohammad Rostami, Spencer Romo, Renhao Xue, Bob Strahan, Diego A. Socolinsky, Boyi Xie, and Md Mofijul Islam. 2026. Can You Trust the Confidence? ConfBench for Vision-Language Models on Document Extraction. *Preprint*, arXiv:2608.01792.

L. S. Shapley. 1953. A value for n-person games. In *Contributions to the Theory of Games (AM-28), Volume II*, pages 307–318. Princeton University Press.

Zhiyun Shi. 2026. VLMs Read Aloud: The Gap Between Recognizing and Using Visual Text. *Preprint*, arXiv:2611.01517.

Qingtao Xia, Siyao Cheng, Jiahua Bao, Jiaxing Du, and Jie Liu. 2026. Found but Not Read: When Extracted Text Closes the Retrieval-Reading Gap in Document Vision-Language Models. *Preprint*, arXiv:2610.02880. ArXiv comment: 5 pages, 5 figures, 3 tables. Submitted to ICASSP 2027.

Fengbin Zhu, Wenqiang Lei, Fuli Feng, Chao Wang, Haozhou Zhang, and Tat-Seng Chua. 2022. Towards

complex document understanding by discrete reasoning. In *Proceedings of the 30th ACM International Conference on Multimedia*, pages 4857–4866.

Fengbin Zhu, Wenqiang Lei, Youcheng Huang, Chao Wang, Shuo Zhang, Jiancheng Lv, Fuli Feng, and Tat-Seng Chua. 2021. TAT-QA: A question answering benchmark on a hybrid of tabular and textual content in finance. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 3277–3287, Online. Association for Computational Linguistics.

Jinguo Zhu, Weiyun Wang, Zhe Chen, Zhaoyang Liu, Shenglong Ye, Lixin Gu, Hao Tian, Yuchen Duan, Weijie Su, Jie Shao, Zhangwei Gao, Erfei Cui, Xuehui Wang, Yue Cao, Yangzhou Liu, Xingguang Wei, Hongjie Zhang, Haomin Wang, Weiye Xu, and 32 others. 2025. InternVL3: Exploring Advanced Training and Test-Time Recipes for Open-Source Multimodal Models. *Preprint*, arXiv:2504.10479.